

A Linear-Algebra Guide to the Strange Life of High-Dimensional Data

Companion Document: Complete Proofs and Derivations

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This document derives, in full, every mathematical claim made in our submission. It contains the mathematics only; figures, code, and numerical pipelines live in the report and notebook. Each part is stated as it appears in the question paper, then proved or derived step by step.

Disclosure on the use of AI. To be fully transparent: the L^AT_EX typesetting of this document was generated by an AI tool. *All of the mathematical proofs and derivations themselves were worked out by hand by the students.* The AI only transcribed our handwritten working into L^AT_EX; it did not originate any of the mathematics. The handwritten originals are the primary evidence of our work and are available here:

[Google Drive folder --- handwritten notes and derivations](#)

1 Markov Matrices and the Fate of a Population

Setup. Two organism types have fractional populations $p_1(t), p_2(t) \in [0, 1]$ evolving in discrete generations $t = 0, 1, 2, \dots$ by $\mathbf{P}(t+1) = \mathbb{M} \mathbf{P}(t)$, where

$$\mathbf{P}(t) = \begin{pmatrix} p_1(t) \\ p_2(t) \end{pmatrix}, \quad \mathbb{M} = \begin{pmatrix} 1 - \varepsilon & 0 \\ \varepsilon & 1 \end{pmatrix}, \quad \varepsilon \in (0, 1). \quad (1)$$

Carrying out the matrix–vector product in (1) gives the two scalar update equations

$$p_1(t+1) = (1 - \varepsilon) p_1(t), \quad p_2(t+1) = \varepsilon p_1(t) + p_2(t). \quad (2)$$

1.1 (a) Conservation of total population

Task. Show that $p_1(t) + p_2(t)$ is constant for every t .

Let $S(t) = p_1(t) + p_2(t)$. Add the two equations of (2):

$$\begin{aligned} S(t+1) &= p_1(t+1) + p_2(t+1) \\ &= (1 - \varepsilon) p_1(t) + [\varepsilon p_1(t) + p_2(t)] \\ &= p_1(t) - \varepsilon p_1(t) + \varepsilon p_1(t) + p_2(t) \\ &= p_1(t) + p_2(t) = S(t). \end{aligned} \quad (3)$$

The ε terms cancel exactly, so $S(t+1) = S(t)$ for every t . By induction on t ,

$$\boxed{S(t) = p_1(t) + p_2(t) = p_1(0) + p_2(0) = S(0) \quad \text{for all } t.} \quad (4)$$

The total population is conserved. (Equivalently: each column of $\mathbb{M}$ sums to 1, so $\mathbf{1}^\top \mathbb{M} = \mathbf{1}^\top$ and $\mathbf{1}^\top \mathbf{P}(t+1) = \mathbf{1}^\top \mathbb{M} \mathbf{P}(t) = \mathbf{1}^\top \mathbf{P}(t)$.)

1.2 (b) Eigenvalues and eigenvectors

Task. Compute the eigenvalues λ_1, λ_2 and eigenvectors $\mathbf{v}_1, \mathbf{v}_2$ of $\mathbb{M}$.

Eigenvalues. The characteristic equation is $\det(\mathbb{M} - \lambda \mathbf{I}) = 0$. Writing the matrix out,

$$\mathbb{M} - \lambda \mathbf{I} = \begin{pmatrix} 1 - \varepsilon - \lambda & 0 \\ \varepsilon & 1 - \lambda \end{pmatrix}, \quad (5)$$

and using $\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$,

$$\begin{aligned} \det(\mathbb{M} - \lambda \mathbf{I}) &= (1 - \varepsilon - \lambda)(1 - \lambda) - (0)(\varepsilon) \\ &= (1 - \varepsilon - \lambda)(1 - \lambda) = 0. \end{aligned} \quad (6)$$

The product vanishes when either factor does, so

$$\boxed{\lambda_1 = 1, \quad \lambda_2 = 1 - \varepsilon.} \quad (7)$$

(Consistent with $\mathbb{M}$ being lower-triangular: its eigenvalues are the diagonal entries $1 - \varepsilon$ and 1 .)

Eigenvector for $\lambda_1 = 1$. Solve $(\mathbb{M} - \mathbf{I})\mathbf{v}_1 = \mathbf{0}$:

$$\mathbb{M} - \mathbf{I} = \begin{pmatrix} -\varepsilon & 0 \\ \varepsilon & 0 \end{pmatrix}, \quad \begin{pmatrix} -\varepsilon & 0 \\ \varepsilon & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -\varepsilon x \\ \varepsilon x \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}. \quad (8)$$

Both rows give $\varepsilon x = 0$, and since $\varepsilon \neq 0$ this forces $x = 0$ with y free. Taking $y = 1$,

$$\boxed{\mathbf{v}_1 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.} \quad (9)$$

Eigenvector for $\lambda_2 = 1 - \varepsilon$. Solve $(\mathbb{M} - (1 - \varepsilon)\mathbf{I})\mathbf{v}_2 = \mathbf{0}$:

$$\mathbb{M} - (1 - \varepsilon)\mathbf{I} = \begin{pmatrix} 0 & 0 \\ \varepsilon & \varepsilon \end{pmatrix}, \quad \begin{pmatrix} 0 & 0 \\ \varepsilon & \varepsilon \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ \varepsilon(x + y) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}. \quad (10)$$

The non-trivial row gives $\varepsilon(x + y) = 0$, hence $x + y = 0$, i.e. $x = -y$. Taking $x = 1$,

$$\boxed{\mathbf{v}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.} \quad (11)$$

$\mathbf{v}_1$ is the all-type-2 state the dynamics never leaves; $\mathbf{v}_2$ is the transient direction along which type 1 decays.

1.3 (c) The initial state in the eigenbasis

Task. With $p_1(0) = \frac{1}{2}$, write $\mathbf{P}(0)$ as a linear combination $c_1\mathbf{v}_1 + c_2\mathbf{v}_2$.

Expand the combination using $\mathbf{v}_1 = (0, 1)^\top$, $\mathbf{v}_2 = (1, -1)^\top$:

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 = c_1 \begin{pmatrix} 0 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} c_2 \\ c_1 - c_2 \end{pmatrix} \stackrel{!}{=} \begin{pmatrix} p_1(0) \\ p_2(0) \end{pmatrix}. \quad (12)$$

Matching components,

$$c_2 = p_1(0), \quad c_1 - c_2 = p_2(0) \implies c_1 = p_1(0) + p_2(0) = S(0). \quad (13)$$

So c_1 is exactly the conserved total of part (a). With the natural normalisation $p_1(0) + p_2(0) = 1$ and $p_1(0) = \frac{1}{2}$ (hence $p_2(0) = \frac{1}{2}$),

$$c_2 = \frac{1}{2}, \quad c_1 = 1, \quad \boxed{\mathbf{P}(0) = \mathbf{v}_1 + \frac{1}{2}\mathbf{v}_2.} \quad (14)$$

1.4 (d) Closed-form time evolution

Task. With $\mathbf{P}(0) = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2$, write $\mathbf{P}(t)$ for $t > 0$ in terms of $c_1, c_2, \mathbf{v}_1, \mathbf{v}_2, \lambda_1, \lambda_2$.

Because $\mathbb{M}\mathbf{v}_i = \lambda_i \mathbf{v}_i$ and $\mathbb{M}$ is linear, apply $\mathbb{M}$ once:

$$\mathbf{P}(1) = \mathbb{M}\mathbf{P}(0) = \mathbb{M}(c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2) = c_1 \mathbb{M}\mathbf{v}_1 + c_2 \mathbb{M}\mathbf{v}_2 = c_1 \lambda_1 \mathbf{v}_1 + c_2 \lambda_2 \mathbf{v}_2. \quad (15)$$

Apply $\mathbb{M}$ again, reusing $\mathbb{M}\mathbf{v}_i = \lambda_i \mathbf{v}_i$:

$$\mathbf{P}(2) = \mathbb{M}\mathbf{P}(1) = c_1 \lambda_1 \mathbb{M}\mathbf{v}_1 + c_2 \lambda_2 \mathbb{M}\mathbf{v}_2 = c_1 \lambda_1^2 \mathbf{v}_1 + c_2 \lambda_2^2 \mathbf{v}_2. \quad (16)$$

Induction. Suppose $\mathbf{P}(t) = c_1 \lambda_1^t \mathbf{v}_1 + c_2 \lambda_2^t \mathbf{v}_2$. Then $\mathbf{P}(t+1) = \mathbb{M}\mathbf{P}(t) = c_1 \lambda_1^t (\lambda_1 \mathbf{v}_1) + c_2 \lambda_2^t (\lambda_2 \mathbf{v}_2) = c_1 \lambda_1^{t+1} \mathbf{v}_1 + c_2 \lambda_2^{t+1} \mathbf{v}_2$, completing the step. Hence $\mathbb{M}^t \mathbf{v}_i = \lambda_i^t \mathbf{v}_i$ and

$$\mathbf{P}(t) = \mathbb{M}^t \mathbf{P}(0) = c_1 \lambda_1^t \mathbf{v}_1 + c_2 \lambda_2^t \mathbf{v}_2. \quad (17)$$

Substituting $\lambda_1 = 1, \lambda_2 = 1 - \varepsilon$,

$$\boxed{\mathbf{P}(t) = c_1 \mathbf{v}_1 + c_2 (1 - \varepsilon)^t \mathbf{v}_2.} \quad (18)$$

The eigenbasis turns repeated matrix multiplication into independent scalar powers: mode i is simply scaled by λ_i^t .

1.5 (e) Long-time behaviour

Task. (i) Show $\mathbf{P}(t)$ approaches the eigenvector with the larger eigenvalue. (ii) Explain how the structure of $\mathbb{M}$ forces this.

(i) The limit. Since $\varepsilon \in (0, 1)$ we have $0 < 1 - \varepsilon < 1$, so $\lambda_2^t = (1 - \varepsilon)^t \rightarrow 0$ as $t \rightarrow \infty$, while $\lambda_1^t = 1^t = 1$. The transient term in (d) vanishes:

$$\mathbf{P}(t) \xrightarrow{t \rightarrow \infty} c_1 \mathbf{v}_1 = (p_1(0) + p_2(0)) \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ p_1(0) + p_2(0) \end{pmatrix}. \quad (19)$$

The limit is a multiple of $\mathbf{v}_1$, the eigenvector with the larger eigenvalue $\lambda_1 = 1$; all mass ends up as type 2.

(ii) Why the structure forces this. The second column of $\mathbb{M}$ is $(0, 1)^\top$, so $\mathbb{M}\mathbf{v}_1 = \mathbf{v}_1$: type 2 is *absorbing*, once an individual is type 2 it never converts. Type 1 satisfies $p_1(t+1) = (1 - \varepsilon)p_1(t)$, i.e. $p_1(t) = (1 - \varepsilon)^t p_1(0) \rightarrow 0$: it loses a fixed fraction ε each step and is never replenished, so it drains to zero. The eigenvalue $\lambda_1 = 1$ carries the surviving steady state and $\lambda_2 = 1 - \varepsilon < 1$ sets the geometric rate at which the rest drains into it.

1.6 (f) Numerical validation of (d) and (e)

Task. Validate (d) and (e) numerically.

The validation is logical rather than analytic: two independently computed trajectories must agree. *Track A* iterates the definition $\mathbf{P}(s+1) = \mathbb{M}\mathbf{P}(s)$ from $\mathbf{P}(0) = (\frac{1}{2}, \frac{1}{2})^\top$ with no analytic input. *Track B* evaluates the closed form $\mathbf{P}(s) = c_1 \mathbf{v}_1 + c_2 (1 - \varepsilon)^s \mathbf{v}_2$ from (d). At $\varepsilon = 0.1$ over $T = 100$ generations,

$$\max_{0 \leq s \leq T} \|\mathbf{P}_A(s) - \mathbf{P}_B(s)\|_\infty \approx 10^{-16}, \quad (20)$$

i.e. agreement to the last bit of double precision, confirming (d). The trajectory settles on $(0, 1)^\top$, confirming (e). On a semi-logarithmic axis the simulated $p_1(s)$ falls exactly on $\frac{1}{2}(1 - \varepsilon)^s$; a straight line there is the signature of geometric decay at rate $1 - \varepsilon$.

1.7 (g) A conversion rate that fades over time

Task. Describe how (d) changes if $\varepsilon(t) = \varepsilon_0 e^{-t/t_0}$ with $\varepsilon_0, t_0 > 0$.

Now the update matrix $\mathbb{M}(t) = \begin{pmatrix} 1-\varepsilon(t) & 0 \\ \varepsilon(t) & 1 \end{pmatrix}$ changes every step. The crucial observation is that the eigenvectors do not depend on ε , so the same $\mathbf{v}_1, \mathbf{v}_2$ diagonalise *every* $\mathbb{M}(t)$:

$$\mathbb{M}(t) \mathbf{v}_1 = \mathbf{v}_1, \quad \mathbb{M}(t) \mathbf{v}_2 = (1 - \varepsilon(t)) \mathbf{v}_2. \quad (21)$$

The state is the ordered product $\mathbf{P}(t) = \mathbb{M}(t-1) \mathbb{M}(t-2) \cdots \mathbb{M}(0) \mathbf{P}(0)$. Acting on $\mathbf{P}(0) = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2$, each factor leaves $\mathbf{v}_1$ fixed and multiplies $\mathbf{v}_2$ by $1 - \varepsilon(s)$ at step s ; these accumulate into a product:

$$\begin{aligned} \mathbf{P}(t) &= c_1 \left[\prod_{s=0}^{t-1} \lambda_1 \right] \mathbf{v}_1 + c_2 \left[\prod_{s=0}^{t-1} (1 - \varepsilon(s)) \right] \mathbf{v}_2 \\ &= \boxed{c_1 \mathbf{v}_1 + c_2 \left[\prod_{s=0}^{t-1} (1 - \varepsilon_0 e^{-s/t_0}) \right] \mathbf{v}_2}. \end{aligned} \quad (22)$$

The geometric power $(1 - \varepsilon)^t$ of part (d) is replaced by a genuine *product* $\prod_s (1 - \varepsilon(s))$; writing $(1 - \varepsilon(t))^t$ would be wrong because the matrix is not fixed.

The product converges to a positive limit. Let $\Pi_t = \prod_{s=0}^{t-1} (1 - \varepsilon(s))$. Since $0 < \varepsilon(s) < 1$, Π_t is positive and decreasing, so $\Pi_t \downarrow \Pi_\infty \geq 0$. Take logarithms:

$$-\ln \Pi_t = \sum_{s=0}^{t-1} -\ln(1 - \varepsilon(s)). \quad (23)$$

For $0 < u < 1$ one has $0 < -\ln(1 - u) \leq \frac{u}{1 - u}$, and with $\varepsilon(s) = \varepsilon_0 e^{-s/t_0} \leq \varepsilon_0$,

$$\sum_{s=0}^{\infty} -\ln(1 - \varepsilon(s)) \leq \frac{1}{1 - \varepsilon_0} \sum_{s=0}^{\infty} \varepsilon_0 e^{-s/t_0} = \frac{\varepsilon_0}{(1 - \varepsilon_0)(1 - e^{-1/t_0})} < \infty, \quad (24)$$

a convergent geometric series. A bounded sum of positive terms converges, so $-\ln \Pi_\infty < \infty$, hence $\Pi_\infty > 0$. Therefore

$$\boxed{\mathbf{P}(\infty) = c_1 \mathbf{v}_1 + c_2 \Pi_\infty \mathbf{v}_2, \quad \Pi_\infty > 0.} \quad (25)$$

Contrast with (d)–(e). With ε constant every factor equals the same number < 1 , so $\Pi_t = (1 - \varepsilon)^t \rightarrow 0$ and type 1 vanishes. With a fading rate the factors $1 - \varepsilon(s) \rightarrow 1$ fast enough that the product freezes at $\Pi_\infty > 0$: the conversion effectively switches off and a permanent residue $c_2 \Pi_\infty$ survives in the $\mathbf{v}_2$ direction. *A fading conversion rate never fully depletes type 1.*

1.8 (h) Validation of the fading-rate result

Task. Validate (g) numerically, as in (f).

Again two tracks: iterate $\mathbf{P}(s+1) = \mathbb{M}(s) \mathbf{P}(s)$ with a freshly built $\mathbb{M}(s)$ at each step ($\varepsilon_0 = 0.5$, $t_0 = 10$), and compare against (22); the maximum discrepancy is again $\approx 10^{-16}$. The trajectory bends away from zero and locks onto the positive plateau $c_2 \Pi_\infty$.

Small-rate (large t_0) estimate of the residue. From the log-sum above, for small $\varepsilon(s)$ the bound $-\ln(1 - u) \approx u$ is tight, so

$$-\ln \Pi_\infty \approx \sum_{s=0}^{\infty} \varepsilon_0 e^{-s/t_0} = \frac{\varepsilon_0}{1 - e^{-1/t_0}} \xrightarrow{t_0 \gg 1} \varepsilon_0 t_0, \quad (26)$$

using $1 - e^{-1/t_0} \approx 1/t_0$. Hence $\Pi_\infty \approx e^{-\varepsilon_0 t_0}$ and the surviving type-1 mass is

$$c_2 \Pi_\infty \approx \frac{1}{2} e^{-\varepsilon_0 t_0}, \quad (27)$$

which matches the measured residue to the expected accuracy.

2 The Power Method and the Singular Value Decomposition

Setup. Let $\mathbb{X}$ be a symmetric $n \times n$ matrix with eigenvalues $|\lambda_1| > |\lambda_2| \geq \dots \geq |\lambda_n|$ and an orthonormal eigenbasis $\mathbf{v}_1, \dots, \mathbf{v}_n$ (so $\mathbf{v}_i \cdot \mathbf{v}_j = \delta_{ij}$). The power method starts from a unit vector $\mathbf{x}_0$ and iterates

$$\mathbf{x}_{k+1} = \frac{\mathbb{X}\mathbf{x}_k}{\|\mathbb{X}\mathbf{x}_k\|}, \quad k = 0, 1, 2, \dots \quad (28)$$

2.1 (a) Convergence to the dominant eigenvector

Claim 2.1. If $\mathbf{x}_0$ has a nonzero component along $\mathbf{v}_1$, then $\mathbf{x}_k \rightarrow \pm \mathbf{v}_1$ as $k \rightarrow \infty$.

Reduce to the unnormalised iterate. Normalisation in (28) only rescales, so $\mathbf{x}_k$ points in the same direction as

$$\mathbf{y}_k := \mathbb{X}^k \mathbf{x}_0 \quad (\mathbf{x}_k = \mathbf{y}_k / \|\mathbf{y}_k\|). \quad (29)$$

A direction is unchanged by positive rescaling, so it suffices to analyse $\mathbf{y}_k$.

Expand in the eigenbasis. Write $\mathbf{x}_0 = \sum_{i=1}^n a_i \mathbf{v}_i$ with $a_1 = \mathbf{x}_0 \cdot \mathbf{v}_1 \neq 0$. Since $\mathbb{X}\mathbf{v}_i = \lambda_i \mathbf{v}_i$ implies $\mathbb{X}^k \mathbf{v}_i = \lambda_i^k \mathbf{v}_i$,

$$\mathbf{y}_k = \mathbb{X}^k \sum_{i=1}^n a_i \mathbf{v}_i = \sum_{i=1}^n a_i \lambda_i^k \mathbf{v}_i = \underbrace{a_1 \lambda_1^k \mathbf{v}_1}_{\text{leading term}} + \underbrace{\sum_{i=2}^n a_i \lambda_i^k \mathbf{v}_i}_{=: \mathbf{r}_k \text{ (residual)}}. \quad (30)$$

Measure the residual against the leading term. Define the ratio

$$\rho_k := \frac{\|\mathbf{r}_k\|}{\|a_1 \lambda_1^k \mathbf{v}_1\|} = \frac{\|\mathbf{r}_k\|}{|a_1| |\lambda_1|^k}. \quad (31)$$

The norm of $\mathbf{r}_k$ follows from taking the dot product with itself and using orthonormality ($\mathbf{v}_i \cdot \mathbf{v}_j = \delta_{ij}$):

$$\mathbf{r}_k \cdot \mathbf{r}_k = \left(\sum_{i \geq 2} a_i \lambda_i^k \mathbf{v}_i \right) \cdot \left(\sum_{j \geq 2} a_j \lambda_j^k \mathbf{v}_j \right) = \sum_{i \geq 2} \sum_{j \geq 2} a_i a_j \lambda_i^k \lambda_j^k (\mathbf{v}_i \cdot \mathbf{v}_j) = \sum_{i \geq 2} a_i^2 \lambda_i^{2k}. \quad (32)$$

Hence $\|\mathbf{r}_k\| = (\sum_{i \geq 2} a_i^2 \lambda_i^{2k})^{1/2}$ and

$$\rho_k = \frac{1}{|a_1|} \sqrt{\sum_{i \geq 2} a_i^2 \left(\frac{\lambda_i}{\lambda_1} \right)^{2k}}. \quad (33)$$

Take the limit. For each $i \geq 2$, $|\lambda_i/\lambda_1| < 1$ (strict, by the spectral-gap assumption $|\lambda_1| > |\lambda_2|$), so $(\lambda_i/\lambda_1)^{2k} \rightarrow 0$. Every term in the finite sum (33) vanishes, giving $\rho_k \rightarrow 0$. Thus the residual becomes negligible relative to the leading term, $\mathbf{y}_k / \|\mathbf{y}_k\| \rightarrow \pm \mathbf{v}_1$, and after normalisation

$$\boxed{\mathbf{x}_k \rightarrow \pm \mathbf{v}_1, \quad \text{sign} = \text{sign}(a_1 \lambda_1^k)}. \quad (34)$$

2.2 (b) Convergence rate and the spectral gap

Task. Show the orthogonal residual $\|\mathbf{r}_k\|$ shrinks, relative to the parallel component $|\mathbf{x}_k \cdot \mathbf{v}_1|$, by at least a factor $|\lambda_2/\lambda_1|$ per iteration.

The relative residual ρ_k of (33) is exactly $\|\mathbf{r}_k\| / |\mathbf{y}_k \cdot \mathbf{v}_1|$ (both numerator and denominator scale the same way under normalisation). Form the consecutive ratio, writing $\omega_i := a_i^2 \lambda_i^{2k} \geq 0$:

$$\frac{\rho_{k+1}}{\rho_k} = \frac{\frac{1}{|a_1| |\lambda_1|^{k+1}} \sqrt{\sum_{i \geq 2} a_i^2 \lambda_i^{2k+2}}}{\frac{1}{|a_1| |\lambda_1|^k} \sqrt{\sum_{i \geq 2} a_i^2 \lambda_i^{2k}}} = \frac{1}{|\lambda_1|} \sqrt{\frac{\sum_{i \geq 2} \omega_i \lambda_i^2}{\sum_{i \geq 2} \omega_i}}. \quad (35)$$

The fraction under the root is a weighted average of the values λ_i^2 ($i \geq 2$) with weights $\omega_i \geq 0$. A weighted average never exceeds the largest value, and the largest is λ_2^2 (since $|\lambda_2| \geq |\lambda_i|$ for $i \geq 2$):

$$\frac{\sum_{i \geq 2} \omega_i \lambda_i^2}{\sum_{i \geq 2} \omega_i} \leq \lambda_2^2. \quad (36)$$

Therefore

$$\boxed{\frac{\rho_{k+1}}{\rho_k} \leq \frac{1}{|\lambda_1|} \sqrt{\lambda_2^2} = \left| \frac{\lambda_2}{\lambda_1} \right| < 1.} \quad (37)$$

Convergence is geometric at rate $|\lambda_2/\lambda_1|$. The controlling quantity is the *spectral gap* between the two largest eigenvalues: a wide gap ($|\lambda_2/\lambda_1| \ll 1$) gives fast, robust convergence; a narrow gap gives slow, fragile convergence.

2.3 (c) When the spectral gap closes

Task. (i) For $\mathbb{X} = \text{diag}(1, -1)$ and generic unit $\mathbf{x}_0 = (a, b)$, compute $\mathbf{x}_k$ for arbitrary k and show it does not converge. (ii) Why is a vanishing gap a fundamental obstruction? (iii) What additional risks arise in finite precision when $|\lambda_2/\lambda_1| \approx 1$?

(i) Explicit non-convergence. Here $|\lambda_1| = |\lambda_2| = 1$, so there is no gap. With $\mathbb{X} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ and $\mathbf{x}_0 = (a, b)^\top$, $a, b \neq 0$,

$$\mathbf{x}_1 = \mathbb{X}\mathbf{x}_0 = \begin{pmatrix} a \\ -b \end{pmatrix}, \quad \mathbf{x}_2 = \mathbb{X}\mathbf{x}_1 = \begin{pmatrix} a \\ b \end{pmatrix}, \quad \mathbf{x}_3 = \mathbb{X}\mathbf{x}_2 = \begin{pmatrix} a \\ -b \end{pmatrix}, \quad \dots \quad (38)$$

so by induction

$$\mathbb{X}^k \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} a \\ (-1)^k b \end{pmatrix}. \quad (39)$$

The norm $\sqrt{a^2 + b^2}$ is preserved at every step, so normalisation does nothing, and the iterate alternates forever:

$$\boxed{\mathbf{x}_k = \frac{1}{\sqrt{a^2 + b^2}} \begin{pmatrix} a \\ (-1)^k b \end{pmatrix} \text{ oscillates between } \frac{(a, b)^\top}{\sqrt{a^2 + b^2}} \text{ and } \frac{(a, -b)^\top}{\sqrt{a^2 + b^2}}.} \quad (40)$$

It never converges to a single vector.

(ii) A structural, not merely slow, failure. The power method works by amplifying each eigenvector by its own factor λ_i^k ; the dominant one wins *only because it is strictly largest in magnitude*, which makes every ratio $(\lambda_i/\lambda_1)^k \rightarrow 0$ in (33). When two leading eigenvalues share a magnitude, $|\lambda_2/\lambda_1| = 1$, the corresponding ratio does not decay: the relative weights of those two modes never change. There is no mechanism to select one of them, so the iterate retains whatever mixture it started with — oscillating when the signs differ (as above), or, if $\lambda_1 = \lambda_2$, sitting at an arbitrary vector of the shared eigenspace. No number of iterations removes this; it is a failure of the method's premise, not of its speed.

(iii) Finite-precision risks near degeneracy. When $|\lambda_2/\lambda_1|$ is close to but below 1, two effects compound. First, the iteration count needed to reach a fixed tolerance scales like $1/(1 - |\lambda_2/\lambda_1|)$ (from the geometric rate of (b)) and blows up. Second, the useful per-step separation between $\mathbf{v}_1$ and $\mathbf{v}_2$ shrinks toward the $\sim 10^{-16}$ rounding error injected by every floating-point operation in double precision. Once the signal-per-step falls to the level of this noise, the computed ordering of λ_1 and λ_2 becomes ambiguous and the returned vector may be an arbitrary combination of $\mathbf{v}_1, \mathbf{v}_2$.

2.4 (d) The symmetric matrices $\mathbb{X}^\top \mathbb{X}$ and $\mathbb{X}\mathbb{X}^\top$

Let $\mathbb{X}$ now be a general $N \times M$ matrix.

(i) Symmetry and positive semi-definiteness

Proposition 2.2. $\mathbb{X}^\top \mathbb{X}$ ($M \times M$) and $\mathbb{X}\mathbb{X}^\top$ ($N \times N$) are symmetric and positive semi-definite; hence each has real, non-negative eigenvalues, and $\sigma_i := \sqrt{\lambda_i}$ is well defined.

Proof. *Symmetry.* Using $(AB)^\top = B^\top A^\top$ and $(A^\top)^\top = A$,

$$(\mathbb{X}^\top \mathbb{X})^\top = \mathbb{X}^\top (\mathbb{X}^\top)^\top = \mathbb{X}^\top \mathbb{X}, \quad (\mathbb{X}\mathbb{X}^\top)^\top = (\mathbb{X}^\top)^\top \mathbb{X}^\top = \mathbb{X}\mathbb{X}^\top. \quad (41)$$

Positive semi-definiteness. For any $\mathbf{w} \in \mathbb{R}^M$, grouping the product as $(\mathbb{X}\mathbf{w})^\top (\mathbb{X}\mathbf{w})$,

$$\mathbf{w}^\top (\mathbb{X}^\top \mathbb{X}) \mathbf{w} = (\mathbb{X}\mathbf{w})^\top (\mathbb{X}\mathbf{w}) = \|\mathbb{X}\mathbf{w}\|^2 \geq 0, \quad (42)$$

and for any $\mathbf{w} \in \mathbb{R}^N$,

$$\mathbf{w}^\top (\mathbb{X}\mathbb{X}^\top) \mathbf{w} = (\mathbb{X}^\top \mathbf{w})^\top (\mathbb{X}^\top \mathbf{w}) = \|\mathbb{X}^\top \mathbf{w}\|^2 \geq 0. \quad (43)$$

A real symmetric matrix has real eigenvalues and an orthonormal eigenbasis (spectral theorem). If $A\mathbf{v} = \lambda\mathbf{v}$ with $\|\mathbf{v}\| = 1$ and A PSD, then $\lambda = \mathbf{v}^\top A\mathbf{v} \geq 0$. Applying this to $A = \mathbb{X}^\top \mathbb{X}$ and $A = \mathbb{X}\mathbb{X}^\top$ gives $\lambda_i \geq 0$, so $\sigma_i = \sqrt{\lambda_i} \in \mathbb{R}$. $\square$

(ii) Shared nonzero eigenvalues

Proposition 2.3. $\mathbb{X}^\top \mathbb{X}$ and $\mathbb{X}\mathbb{X}^\top$ have the same nonzero eigenvalues; their common nonzero values are the σ_i^2 .

Proof. Let $\mathbb{X}^\top \mathbb{X} \mathbf{v} = \lambda \mathbf{v}$ with $\lambda \neq 0$ (so $\mathbf{v} \neq \mathbf{0}$). Left-multiply by $\mathbb{X}$ and regroup:

$$\mathbb{X}(\mathbb{X}^\top \mathbb{X} \mathbf{v}) = \mathbb{X}(\lambda \mathbf{v}) \implies (\mathbb{X}\mathbb{X}^\top)(\mathbb{X}\mathbf{v}) = \lambda(\mathbb{X}\mathbf{v}). \quad (44)$$

We must check $\mathbb{X}\mathbf{v} \neq \mathbf{0}$: if $\mathbb{X}\mathbf{v} = \mathbf{0}$ then $\mathbb{X}^\top \mathbb{X} \mathbf{v} = \mathbb{X}^\top \mathbf{0} = \mathbf{0} = \lambda \mathbf{v}$, forcing $\lambda = 0$, a contradiction. Hence $\mathbb{X}\mathbf{v}$ is an eigenvector of $\mathbb{X}\mathbb{X}^\top$ with the same eigenvalue λ . The reverse direction is symmetric: if $\mathbb{X}\mathbb{X}^\top \mathbf{u} = \mu \mathbf{u}$ with $\mu \neq 0$, then $\mathbb{X}^\top \mathbf{u} \neq \mathbf{0}$ and $(\mathbb{X}^\top \mathbb{X})(\mathbb{X}^\top \mathbf{u}) = \mu(\mathbb{X}^\top \mathbf{u})$. So the nonzero spectra coincide; by the definition $\sigma_i = \sqrt{\lambda_i}$ these shared values are the σ_i^2 . $\square$

2.5 (e) A power-method SVD, with a worked example

Task. (i) Give language-agnostic pseudocode returning $\sigma_1, \mathbf{v}_1, \mathbf{u}_1$ for an $N \times M$ matrix, using only the power method and matrix–vector products; justify the choice of $\mathbb{X}^\top \mathbb{X}$ vs $\mathbb{X}\mathbb{X}^\top$. (ii) Apply it by hand to a 3×2 matrix and verify the SVD relations.

(i) The algorithm

The power method needs a symmetric matrix, so from rectangular $\mathbb{X} \in \mathbb{R}^{N \times M}$ build one square of part (d). **Cost argument:** $\mathbb{X}^\top \mathbb{X}$ is $M \times M$ and $\mathbb{X}\mathbb{X}^\top$ is $N \times N$; each power-method step is one matrix–vector product, costing $O(d^2)$ for a $d \times d$ matrix. Choose the *smaller* square, $d = \min(N, M)$: use $\mathbb{X}\mathbb{X}^\top$ when $N \ll M$, and $\mathbb{X}^\top \mathbb{X}$ when $M \ll N$. The companion vector follows from one extra product via $\mathbf{u}_1 = \mathbb{X}\mathbf{v}_1/\sigma_1$ (or $\mathbf{v}_1 = \mathbb{X}^\top \mathbf{u}_1/\sigma_1$).

Leading singular triplet $(\sigma_1, \mathbf{v}_1, \mathbf{u}_1)$ of $\mathbb{X} \in \mathbb{R}^{N \times M}$.

input: matrix $\mathbb{X}$; tolerance $\tau > 0$; iteration cap $K_{\max}$

output: $\sigma_1 \geq 0$; right vector $\mathbf{v}_1 \in \mathbb{R}^M$; left vector $\mathbf{u}_1 \in \mathbb{R}^N$

1. **if** $N \geq M$ **then** $A \leftarrow \mathbb{X}^T \mathbb{X}$ **else** $A \leftarrow \mathbb{X} \mathbb{X}^T$ (A is $d \times d$, $d = \min(N, M)$)
2. choose unit $\mathbf{q} \in \mathbb{R}^d$, $\|\mathbf{q}\| = 1$, not orthogonal to the dominant eigenvector
3. **for** $k = 1, \dots, K_{\max}$:
4. $\mathbf{w} \leftarrow A \mathbf{q}$; $\mathbf{q}' \leftarrow \mathbf{w} / \|\mathbf{w}\|$ (*one matrix–vector product, then renormalise*)
5. **if** $\|\mathbf{q}' - \mathbf{q}\| < \tau$ **then** $\mathbf{q} \leftarrow \mathbf{q}'$; **break**
6. $\mathbf{q} \leftarrow \mathbf{q}'$
7. $\lambda_1 \leftarrow \mathbf{q}^T A \mathbf{q}$; $\sigma_1 \leftarrow \sqrt{\lambda_1}$ (*Rayleigh quotient; A is PSD so $\lambda_1 \geq 0$*)
8. **if** $A = \mathbb{X}^T \mathbb{X}$ **then** $\mathbf{v}_1 \leftarrow \mathbf{q}$, $\mathbf{u}_1 \leftarrow \mathbb{X} \mathbf{v}_1 / \sigma_1$
9. **else** $\mathbf{u}_1 \leftarrow \mathbf{q}$, $\mathbf{v}_1 \leftarrow \mathbb{X}^T \mathbf{u}_1 / \sigma_1$
10. **return** $\sigma_1, \mathbf{v}_1, \mathbf{u}_1$

(ii) Worked example by hand

Take

$$\mathbb{X} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}, \quad N = 3, M = 2. \quad (45)$$

Since $N \geq M$, line 1 forms the smaller square $A = \mathbb{X}^T \mathbb{X}$:

$$\mathbb{X}^T \mathbb{X} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}. \quad (46)$$

Eigenvalues. $\det(\mathbb{X}^T \mathbb{X} - \lambda \mathbf{I}) = 0$:

$$\det \begin{pmatrix} 2 - \lambda & 1 \\ 1 & 2 - \lambda \end{pmatrix} = (2 - \lambda)^2 - 1 = 0 \implies \lambda^2 - 4\lambda + 3 = 0 \implies (\lambda - 1)(\lambda - 3) = 0, \quad (47)$$

so $\lambda \in \{1, 3\}$ and the dominant one is $\lambda_1 = 3$, giving $\sigma_1 = \sqrt{3}$.

Right singular vector. Solve $(\mathbb{X}^T \mathbb{X} - 3\mathbf{I})\mathbf{v}_1 = \mathbf{0}$:

$$\begin{pmatrix} 2 - 3 & 1 \\ 1 & 2 - 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -x + y \\ x - y \end{pmatrix} = \mathbf{0} \implies x = y. \quad (48)$$

Normalising ($x = y, x^2 + y^2 = 1$) gives

$$\mathbf{v}_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}. \quad (49)$$

Left singular vector.

$$\mathbf{u}_1 = \frac{\mathbb{X} \mathbf{v}_1}{\sigma_1} = \frac{1}{\sqrt{3}} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}. \quad (50)$$

Verification of both defining relations.

$$\mathbb{X} \mathbf{v}_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} = \sqrt{3} \cdot \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} = \sigma_1 \mathbf{u}_1, \quad (51)$$

$$\mathbb{X}^T \mathbf{u}_1 = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} = \frac{1}{\sqrt{6}} \begin{pmatrix} 3 \\ 3 \end{pmatrix} = \frac{3}{\sqrt{6}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \sqrt{3} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \sigma_1 \mathbf{v}_1, \quad (52)$$

using $3/\sqrt{6} = \sqrt{6}/2 = \sqrt{3}/\sqrt{2}$. Both relations hold, confirming the triplet

$$(\sigma_1, \mathbf{v}_1, \mathbf{u}_1) = \left(\sqrt{3}, \frac{1}{\sqrt{2}}(1, 1)^T, \frac{1}{\sqrt{6}}(1, 1, 2)^T \right). \quad (53)$$

3 Design, Covariance, and Gram Matrices

Setup. The design matrix $\mathbb{X} \in \mathbb{R}^{N \times M}$ has rows indexing samples (vials) and columns indexing features (frequency channels); $\mathbb{X}_{ij}$ is the absorbance of vial i at frequency j . The supplied array `mystery_matrix1.npy` loads as 3401×1000 : it is stored transposed, so we transpose to put samples in rows, and we use its true feature count $M = 3401$ (the prompt’s 3041 is a digit transposition). After cleaning, $\tilde{N}$ samples remain.

3.1 (a) Cleaning by the row-norm

Task. Identify and remove the T corrupted (empty) measurements, and find a second error.

The relevant quantity is the Euclidean norm of each spectrum,

$$\|\mathbf{x}_i\| = \left(\sum_{j=1}^M \mathbb{X}_{ij}^2 \right)^{1/2}. \quad (54)$$

An empty vial records only small, flat background noise, hence small norm. Sorting the 1000 norms exposes the structure: 40 vials at $\|\mathbf{x}\| \approx 1.6$, then a clean gap, then the bulk from $\|\mathbf{x}\| \approx 4.3$. Thresholding in the gap flags exactly $T = 40$ empties (rows 960–999); the count is invariant for any cutoff in $[2.0, 4.0]$, so it is principled rather than tuned. A second defect sits in the upper tail: 36 vials at $\|\mathbf{x}\| \approx 8.9$ (rows 924–959), separated by a gap $\approx 5\times$ wider than any internal gap. We retain sample i iff

$$\tau_{\text{lo}} \leq \|\mathbf{x}_i\| \leq \tau_{\text{hi}}, \quad (55)$$

implemented as a Boolean keep-mask (non-destructive, index-preserving), leaving

$$\tilde{N} = 1000 - 40 - 36 = 924 \text{ spectra.} \quad (56)$$

(The 36 form a single isolated cluster in the leading-PC plane, part (e); removing them drops the cluster count by exactly one, confirming they are one corrupted group, not a distinct recipe.)

3.2 (b) Feature variance and the covariance matrix

Task. Mean-centre the data; identify the highest-variance feature; relate the covariance diagonal to variance and the trace to total variance.

Mean-centring. For each feature j compute the mean over the $\tilde{N}$ retained samples and subtract it:

$$\bar{\mathbb{X}}_j = \frac{1}{\tilde{N}} \sum_{i=1}^{\tilde{N}} \mathbb{X}_{ij}, \quad \tilde{\mathbb{X}}_{ij} = \mathbb{X}_{ij} - \bar{\mathbb{X}}_j, \quad (57)$$

so each column of $\tilde{\mathbb{X}}$ has zero mean: $\sum_i \tilde{\mathbb{X}}_{ij} = 0$. The covariance matrix is

$$\mathbb{C} = \frac{1}{\tilde{N}} \tilde{\mathbb{X}}^T \tilde{\mathbb{X}} \in \mathbb{R}^{M \times M}, \quad \mathbb{C}_{jk} = \frac{1}{\tilde{N}} \sum_{i=1}^{\tilde{N}} \tilde{\mathbb{X}}_{ij} \tilde{\mathbb{X}}_{ik}. \quad (58)$$

Proposition 3.1. $\mathbb{C}$ is symmetric, and its diagonal entries are the feature variances, $\mathbb{C}_{jj} = \frac{1}{\tilde{N}} \sum_i \tilde{\mathbb{X}}_{ij}^2 = \text{Var}_j$.

Proof. $\mathbb{C}^T = \frac{1}{\tilde{N}} (\tilde{\mathbb{X}}^T \tilde{\mathbb{X}})^T = \frac{1}{\tilde{N}} \tilde{\mathbb{X}}^T \tilde{\mathbb{X}} = \mathbb{C}$. Setting $k = j$, $\mathbb{C}_{jj} = \frac{1}{\tilde{N}} \sum_i \tilde{\mathbb{X}}_{ij}^2$. Since column j has zero mean, this is precisely the variance of feature j across samples, $\text{Var}_j = \frac{1}{\tilde{N}} \sum_i (\mathbb{X}_{ij} - \bar{\mathbb{X}}_j)^2$. $\square$

Proposition 3.2. The trace $\text{tr } \mathbb{C} = \sum_j \text{Var}_j$ is the total variance; it is invariant under any orthonormal change of basis, and equals $\sum_i \lambda_i$.

Proof. $\text{tr } \mathbb{C} = \sum_j \mathbb{C}_{jj} = \sum_j \text{Var}_j$. For orthonormal Q ($Q^\top Q = I$), the cyclic property of the trace gives $\text{tr}(Q^\top \mathbb{C} Q) = \text{tr}(\mathbb{C} Q Q^\top) = \text{tr } \mathbb{C}$. By the spectral theorem $\mathbb{C} = Q \Lambda Q^\top$ with Λ diagonal of eigenvalues, so $\text{tr } \mathbb{C} = \text{tr}(Q \Lambda Q^\top) = \text{tr } \Lambda = \sum_i \lambda_i$. $\square$

Numerical results. $\text{tr } \mathbb{C} \approx 7.90$; the single most variable feature is $j^\star = 910$ with $\mathbb{C}_{j^\star j^\star} \approx 0.0199$. The variance profile is not spread evenly but concentrates in a few narrow molecular bands, with flat baseline between — the data is highly redundant.

3.3 (c) The eigenspectrum (PCA)

Task. Report λ_1/λ_2 ; identify the elbow $k^\star$; say what a large ratio means.

Variance along a direction (Rayleigh quotient). For a unit $\mathbf{w} \in \mathbb{R}^M$, the projected scores are $z_i = \tilde{\mathbf{x}}_i \cdot \mathbf{w}$. Their variance (the centred data has zero mean, so the scores do too) is

$$\frac{1}{\bar{N}} \sum_{i=1}^{\bar{N}} z_i^2 = \frac{1}{\bar{N}} \sum_i (\mathbf{w}^\top \tilde{\mathbf{x}}_i) (\tilde{\mathbf{x}}_i^\top \mathbf{w}) = \mathbf{w}^\top \left(\frac{1}{\bar{N}} \sum_i \tilde{\mathbf{x}}_i \tilde{\mathbf{x}}_i^\top \right) \mathbf{w} = \mathbf{w}^\top \left(\frac{1}{\bar{N}} \tilde{\mathbf{X}}^\top \tilde{\mathbf{X}} \right) \mathbf{w} = \mathbf{w}^\top \mathbb{C} \mathbf{w}. \quad (59)$$

For a unit eigenvector $\mathbf{w} = \mathbf{v}_k$ with $\mathbb{C} \mathbf{v}_k = \lambda_k \mathbf{v}_k$,

$$\mathbf{v}_k^\top \mathbb{C} \mathbf{v}_k = \mathbf{v}_k^\top (\lambda_k \mathbf{v}_k) = \lambda_k \|\mathbf{v}_k\|^2 = \lambda_k, \quad (60)$$

so the eigenvalue λ_k is the variance captured along $\mathbf{v}_k$. Moreover $\mathbf{w}^\top \mathbb{C} \mathbf{w} = \frac{1}{\bar{N}} \|\tilde{\mathbf{X}} \mathbf{w}\|^2 \geq 0$, so $\mathbb{C}$ is PSD and every $\lambda_i \geq 0$.

Diagonalisation. By the spectral theorem $\mathbb{C}$ has an orthonormal eigenbasis; eigenvalues are sorted decreasingly and rounding-level negatives clipped to 0. The leading eigenvalues are

k	1	2	3	4	5	6	7	8	9
λ_k	2.492	0.705	0.590	0.254	0.123	0.105	0.078	0.039	0.0093

The dominance ratio is

$$\frac{\lambda_1}{\lambda_2} \approx \frac{2.492}{0.705} \approx 3.53, \quad \frac{\lambda_1}{\text{tr } \mathbb{C}} \approx \frac{2.492}{7.90} \approx 31\%. \quad (61)$$

A large ratio means one direction dominates the variation: the data has strong low-dimensional structure, and (by Question 2) the wide spectral gap would make the power method converge quickly here.

The elbow. The signal/noise boundary is multiplicative on the logarithmic scale, so we locate the elbow by the largest *multiplicative* drop between consecutive eigenvalues,

$$k^\star = \arg \max_k \frac{\lambda_k}{\lambda_{k+1}}, \quad \frac{\lambda_8}{\lambda_9} \approx \frac{0.039}{0.0093} \approx 4.1, \quad \implies \boxed{k^\star = 8}. \quad (62)$$

Eight signal components sit above a flat noise floor of ~ 900 tiny, slowly decreasing eigenvalues; this echoes the ~ 10 dominant molecules of part (h). As a check, $\sum_i \lambda_i \approx \text{tr } \mathbb{C} \approx 7.90$ (Prop. 3.2).

Why not the “95%” rule. The eight signal components explain only $\approx 55\%$ of the variance; reaching 95% requires 667 components, because the noise floor, negligible term by term, is spread over hundreds of dimensions:

cumulative variance	50%	80%	90%	95%	99%
components k	4	309	516	667	850

A 95% cut would drag in ≈ 660 noise directions and overstate the dimensionality; the elbow is the honest measure.

Threshold-free corroboration. Define the explained-variance ratio and the participation ratio (a continuous effective dimension),

$$r_k = \frac{\lambda_k}{\sum_j \lambda_j}, \quad d_{\text{eff}} = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2}. \quad (63)$$

By Cauchy–Schwarz, $1 \leq d_{\text{eff}} \leq \text{rank}(\mathbb{C})$, with d_{eff} near 1 when one eigenvalue dominates and large when many share the variance. Here $d_{\text{eff}} \approx 8.7$, agreeing with the elbow $k^* = 8$ and the ~ 10 molecules: three independent criteria (the multiplicative drop, the participation ratio, and the chemistry) converge on the same handful of dimensions.

3.4 (d) The leading eigenvector

Task. Interpret $\mathbf{v}_1$ as a function of frequency; does it resemble a raw spectrum?

$\mathbf{v}_1 \in \mathbb{R}^M$ assigns a loading to each frequency; the scores $z_i = \tilde{\mathbf{x}}_i \cdot \mathbf{v}_1$ are the one-dimensional summaries of greatest spread (the $k = 1$ case of part (c)). Eigenvectors are defined only up to sign, fixed here by making the largest loading positive. The loadings concentrate on two narrow bands ($j \approx 900\text{--}960$ and $3150\text{--}3260$), with $\approx 27\%$ negative entries, so $\mathbf{v}_1$ is a *contrast* direction (a balance of absorption between bands), not overall intensity. Using the scale-invariant cosine similarity

$$\cos(\mathbf{a}, \mathbf{b}) = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|}, \quad (64)$$

$\mathbf{v}_1$ matches a mean-centred spectrum at $\cos \approx 0.92$, but only 0.67 against a raw spectrum and 0.03 against the mean spectrum. Living in the centred space by construction, $\mathbf{v}_1$ is nearly orthogonal to the shared mean and behaves like a *difference spectrum*: it captures how the most distinctive vial departs from the average.

3.5 (e) Clustering in covariance space

Task. Project onto the top four eigenvectors and count the clusters.

Stack the top four eigenvectors as columns of $\mathbb{V}_4 = [\mathbf{v}_1 \ \mathbf{v}_2 \ \mathbf{v}_3 \ \mathbf{v}_4] \in \mathbb{R}^{M \times 4}$. The compressed coordinates are

$$\mathbf{Z} = \tilde{\mathbf{X}} \mathbb{V}_4 \in \mathbb{R}^{\tilde{N} \times 4}, \quad \mathbf{z}_i = (\tilde{\mathbf{x}}_i \cdot \mathbf{v}_1, \tilde{\mathbf{x}}_i \cdot \mathbf{v}_2, \tilde{\mathbf{x}}_i \cdot \mathbf{v}_3, \tilde{\mathbf{x}}_i \cdot \mathbf{v}_4)^T. \quad (65)$$

Cluster counting uses a leader rule with radius r : scanning the points, start a new cluster centre whenever a point is farther than r from every existing centre, i.e. open a new centre at $\mathbf{z}_i$ iff

$$\min_{\text{centres } \mathbf{c}} \|\mathbf{z}_i - \mathbf{c}\| > r. \quad (66)$$

This returns 25 clusters, stable for any r between ≈ 6 and ≈ 12 times the median nearest-neighbour spacing — a property of the data, not the threshold. Each cluster is a set of vials with nearly identical spectra, i.e. one chemical mixture, so the 924 vials are ≈ 25 recipes, well under the “fewer than 100” of part (h). Although $\mathbf{v}_1, \dots, \mathbf{v}_4$ live in feature space, projecting samples onto them exposes sample-space structure, because similar spectra get similar scores.

3.6 (f) The same question through the Gram matrix

Task. Compute the top four eigenvectors of the Gram matrix, project onto them, and count clusters.

The Gram matrix lives in sample space:

$$\mathbb{G} = \frac{1}{\tilde{N}} \tilde{\mathbf{X}} \tilde{\mathbf{X}}^T \in \mathbb{R}^{\tilde{N} \times \tilde{N}}, \quad \mathbb{G}_{ij} = \frac{1}{\tilde{N}} \sum_{k=1}^M \tilde{\mathbf{x}}_{ik} \tilde{\mathbf{x}}_{jk}, \quad (67)$$

so $\mathbb{G}_{ij}$ is the (scaled) dot-product similarity between vials i and j . Because $\tilde{N} = 924 \ll M = 3401$, diagonalising the $\tilde{N} \times \tilde{N}$ Gram matrix instead of the $M \times M$ covariance matrix costs about

$$\left(\frac{M}{\tilde{N}}\right)^2 \approx \left(\frac{3401}{924}\right)^2 \approx 13.5 \quad (68)$$

times less — exactly the “use the smaller square” rule of Question 2(e). By Proposition 2.3 (with $\tilde{\mathbb{X}}$ in place of $\mathbb{X}$) $\mathbb{G}$ shares the nonzero eigenvalues of $\mathbb{C}$; numerically the top eight agree to within 10^{-6} . A Gram eigenvector’s components are themselves the sample coordinates (no projection of $\tilde{\mathbb{X}}$ needed); scaling by $\sqrt{\tilde{N}\lambda}$ matches the units of part (e). The six pairwise Gram scatter plots reproduce those of part (e) up to per-axis sign flips, and the same leader rule again returns 25 clusters.

3.7 (g) Why the two clusterings coincide

Task. Do the covariance and Gram routes give the same clusters? Explain.

Shared spectrum (direct proof).

Proposition 3.3. *If $\mathbb{C}\mathbf{v} = \lambda\mathbf{v}$ with $\mathbb{C} = \frac{1}{N}\tilde{\mathbb{X}}^T\tilde{\mathbb{X}}$, then $\tilde{\mathbb{X}}\mathbf{v}$ is an eigenvector of $\mathbb{G} = \frac{1}{N}\tilde{\mathbb{X}}\tilde{\mathbb{X}}^T$ with the same eigenvalue λ .*

Proof. Compute, regrouping the product:

$$\mathbb{G}(\tilde{\mathbb{X}}\mathbf{v}) = \frac{1}{N}\tilde{\mathbb{X}}\tilde{\mathbb{X}}^T\tilde{\mathbb{X}}\mathbf{v} = \tilde{\mathbb{X}}\left(\frac{1}{N}\tilde{\mathbb{X}}^T\tilde{\mathbb{X}}\right)\mathbf{v} = \tilde{\mathbb{X}}(\mathbb{C}\mathbf{v}) = \tilde{\mathbb{X}}(\lambda\mathbf{v}) = \lambda(\tilde{\mathbb{X}}\mathbf{v}). \quad (69)$$

□

The clean statement via the SVD. Write the centred data as $\tilde{\mathbb{X}} = \mathbf{U}\Sigma\mathbf{V}^T$ (singular value decomposition), where $\mathbf{V} = [\mathbf{v}_1 \ \mathbf{v}_2 \ \dots]$ holds the eigenvectors of $\tilde{\mathbb{C}} = \tilde{\mathbb{X}}^T\tilde{\mathbb{X}}$ and $\Sigma = \text{diag}(\sigma_1, \sigma_2, \dots)$ with $\sigma_i^2 = \tilde{N}\lambda_i$. The covariance scores and the Gram scores are then the *same* matrix:

$$\mathbf{Z} = \tilde{\mathbb{X}}\mathbf{V} = \mathbf{U}\Sigma\mathbf{V}^T\mathbf{V} = \mathbf{U}\Sigma, \quad \mathbf{Z}_g = \mathbf{U}\Sigma = \mathbf{U}\sqrt{\tilde{N}\Lambda}, \quad (70)$$

using $\mathbf{V}^T\mathbf{V} = \mathbf{I}$ on the left, and on the right that the Gram eigenvectors are the columns of $\mathbf{U}$ scaled by $\sigma_i = \sqrt{\tilde{N}\lambda_i}$. Hence

$$\boxed{\mathbf{Z} = \mathbf{Z}_g = \mathbf{U}\Sigma \quad (\text{up to the arbitrary sign of each column}).} \quad (71)$$

Because the two point clouds are identical up to reflecting individual axes, their clusters — and the count — must coincide; both give 25. Numerically the 4×4 correlation matrix between the two score sets is the identity in absolute value, and the coordinates agree in magnitude to $\approx 10^{-14}$ (axes 1 and 4 happen to come out sign-flipped). Covariance space asks *which combination of frequencies* a vial expresses; Gram space asks *which other vials* it resembles; both place vials with nearly identical spectra at the same point.

3.8 (h) The forgetful taste tester

Task. (i) How many unique food samples were there? (ii) Which earlier parts were useful?

(i) The count. After removing the 40 empty and 36 corrupted vials (part a), the 924 genuine spectra fall into 25 tight clusters, each a set of near-identical spectra, i.e. one recipe:

$$\boxed{\approx 25 \text{ unique samples.}} \quad (72)$$

The estimate is stable, not fitted: it holds across clustering radii spanning a factor of two and is reproduced by two independent constructions, the covariance and Gram matrices (part g), whose agreement is guaranteed

by the $\mathbb{X}^T \mathbb{X} / \mathbb{X} \mathbb{X}^T$ duality of Question 2(d). It sits well under the stated bound of 100. The ~ 10 dominant molecules surface as the $k^* = 8$ signal dimensions of the eigenspectrum: each spectrum is a non-negative mixture of ~ 10 absorption signatures, so the centred data needs only that many independent directions. The cluster sizes are near-uniform,

$$24 \times 37 + 1 \times 36 = 924, \quad (73)$$

the fingerprint of a designed taste test measuring a fixed menu of recipes several dozen times each.

(ii) Which parts contributed. The cleaning of (a) was a precondition (outliers would invent spurious clusters); the mean-centring/covariance of (b) and the eigenspectrum of (c) compressed 3401 noisy features to the 8 carrying signal; the projection and clustering of (e) produced the count, the Gram route of (f) reproduced it, and the comparison in (g) turned that agreement into a genuine cross-check resting on the duality proved in Question 2(d). Even Question 1 supplies the guiding idea: the leading eigenvectors of the right matrix carry the structure a system settles into.

4 Supporting Result: Optimality of the PCA Truncation

The compression in Question 3 is not ad hoc but optimal, by the Eckart–Young theorem.

Theorem 4.1 (Eckart–Young, in PCA form). *Let $\tilde{\mathbb{X}} = \sum_{i=1}^r \sigma_i \mathbf{u}_i \mathbf{v}_i^T$ be the SVD of the centred data, with $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0$. Among all matrices of rank at most k , the truncation $\tilde{\mathbb{X}}_k = \sum_{i=1}^k \sigma_i \mathbf{u}_i \mathbf{v}_i^T$ (equivalently, the projection of $\tilde{\mathbb{X}}$ onto its top- k right singular vectors) minimises the squared Frobenius error, and*

$$\min_{\text{rank}(B) \leq k} \left\| \tilde{\mathbb{X}} - B \right\|_F^2 = \left\| \tilde{\mathbb{X}} - \tilde{\mathbb{X}}_k \right\|_F^2 = \sum_{i>k} \sigma_i^2 = \tilde{N} \sum_{i>k} \lambda_i. \quad (74)$$

Consequence for $k^* = 8$. Keeping the 8 signal components replaces each 3401-number spectrum by 8 scores — a $3401/8 \approx 425$ -fold reduction — while retaining the $\approx 55\%$ of the variance that genuinely separates the mixtures. The discarded $\approx 45\%$ is the noise floor, spread as $\approx 0.05\%$ per direction over ~ 900 dimensions, so no single dropped component carries recoverable structure. This is why a handful of directions recovers the 25 recipes from a nominally 3401-dimensional measurement.

End of companion. Every boxed result above is used verbatim in the submission report `SIMC10_PSBK-KKN.pdf`; the numerical values reproduce from the notebook `main.ipynb`.